Lesson Plan — Electrostatics (Electric charge, Coulomb's law, electric field, Gauss's law)

Physics | Grade 11-12 | Foundational | Electric Charges and Fields

Overview

Subject	Physics
Grade band	11-12
Topic	Electrostatics (Electric charge, Coulomb's law, electric field, Gauss's law)
Chapter	Electric Charges and Fields
Source document	1.4.1 Additivity of charges
Periods	6 x 40 min

Learning Objectives

- [remember] State the operational definition of electric field as force per unit positive test charge and give its SI unit.
- [apply] Calculate the magnitude and direction of the electrostatic force between two point charges using Coulomb's law for given q1, q2 and separation r.
- [apply] Use the relation E(r) = (1/4πε0) Q / r^2 r̂ to compute the electric field produced by a point charge Q at a distance r and indicate its direction for positive and negative Q.
- [analyze] Use Gauss's law to derive the electric field of (a) a uniformly charged infinite plane, (b) an infinitely long straight charged wire, and (c) a uniformly charged thin spherical shell for interior and exterior points.
- [understand] Explain that electric charge is quantised in integer multiples of e and that total charge is conserved in an isolated system, citing the microscopic origin (electron/proton) given in the text.
- [apply] Given a dipole moment p and a uniform electric field E, determine the torque τ = p × E and state whether a net force acts on the dipole.

Period-by-Period Plan

Period 1: Charges, Conductors, and Coulomb Force

Concepts	Electrostatics, Everyday examples of static electricity, Two kinds of electric charge and their interaction, Conductors and insulators
Objectives	Calculate the magnitude and direction of the electrostatic force between two point charges using Coulomb's law for given q1, q2 and separation r.
Time allocation	Activate prior knowledge with demonstrations of everyday static electricity (sparks, charged plastic, hair standing); ask students to describe observed forces and record qualitative direction and magnitude impressions in words and simple sketches (anchor in volts/charge verbally) (5 min),

	Introduce the concept of two kinds of charge and conductor vs insulator: define 'charge' with units coulomb (C), show small charged pith-ball/commercial electroscope demo and explain charge redistribution on conductors vs fixed charge on insulators (10 min), Derive Coulomb's law step by step from an experimental setup: explain point-charge idealisation, present empirical 1/r^2 dependence from two-charge experiment, introduce constant k and write magnitude form F = k|q1 q2|/r^2 with units (N), and show vector direction rule (repel/attract) tied to sign (12 min), Guided practice: pairs calculate magnitude and direction of force for two given point charges (explicit numbers, carry units through: q1 in C, q2 in C, r in m F in N), teacher circulates and checks unit algebra and vector direction (8 min), Check understanding and consolidate: short written diagnostic (2 problems: sign/direction and a units check) followed by whole-class discussion linking results to next period's focus on charge quantisation and superposition (5 min)
Why this order	As the opening period, this lesson introduces the two kinds of electric charge, everyday static-electricity examples, and conductor/insulator behaviour; it builds the concrete ideas and measurement conventions needed to derive and apply Coulomb's law and supports the next period on Quantisation of charge, Coulomb's law, and Principle of superposition.

Period 2: Charge Quantisation and Coulomb's Law

Concepts	Quantisation of electric charge, Coulomb's law, Principle of superposition
Objectives	Calculate the magnitude and direction of the electrostatic force between two point charges using Coulomb's law for given q1, q2 and separation r.; Explain that electric charge is quantised in integer multiples of e and that total charge is conserved in an isolated system, citing the microscopic origin (electron/proton) given in the text.
Time allocation	Activate prior knowledge: rapid warm-up recalling conductors/insulators, +/− charge examples, and one schematic of two point charges from the previous lesson (students state expected force direction) (5 min), Introduce quantisation of charge: stepwise explanation linking atoms, electrons (charge $-e = 1.60\times10^{-19}$ C) and protons, show conservation in an isolated system using a simple electron-transfer example with units carried through (7 min), Derive Coulomb's law experimentally and mathematically: present experimental evidence for inverse-square dependence, then derive vector form $F = k\, q1\, q2\, \hat{r} / r^2$ showing units (N) step by step and anchoring to a concrete example (two 1.0 μC charges separated by 0.10 m) (10 min), Demonstration/measurement: teacher demo or video of a torsion-balance style set-up (or simulation) measuring force vs distance, students extract numerical values and check units (N) and proportionality to $q1q2/r^2$ (6 min), Guided practice: pairs solve two worked problems — (a) compute magnitude and direction of force between specified q1, q2, r with units; (b) compute net force from three charges using superposition (students write vector sum steps and units) (8 min), Check and consolidate: short 5-minute quiz of 2 items (one

calculation, one conceptual on quantisation/conservation) followed by whole-class review of answers and statement of how this enables 'Electric field, Test charge and source charge' (4 min)

Why this order	This period builds on 'Charges, Conductors, and Coulomb Force' by formalising charge as quantised (multiples of e) and deriving Coulomb's inverse-square force law and the superposition rule; it prepares students to use point-source forces to define the 'Electric field' and to analyse test/source charge setups in the following period.

Period 3: Defining and Calculating Electric Field

Concepts	Electric field, Electric field, Test charge and source charge
Objectives	State the operational definition of electric field as force per unit positive test charge and give its SI unit.; Use the relation $E(r) = (1/4\pi\varepsilon 0)\ Q / r^2\ \hat{r}$ to compute the electric field produced by a point charge Q at a distance r and indicate its direction for positive and negative Q.
Time allocation	Activate prior knowledge: quick review of Coulomb's law and charge sign (talk and 2 quick questions), recalling units (N, C) and inverse-square behaviour (6 min), Introduce electric field concept: derive $E = F/q_{test}$ from Coulomb's law step-by-step using a point charge Q and a +1.0 μC test charge at distance r (show units N/C and N·m/C where they arise) (10 min), Demonstration / measurement: instructor-led desk demo using two charged spheres and a small test probe (qualitative needle deflection or force sensor), record measured force (N) and compute E (N/C) at measured r (m) (8 min), Guided practice: pairs solve two worked examples—compute magnitude and direction of E for Q = +5.0 μC and Q = −3.0 μC at given r values (carry units N/C and show $\hat{r}$ direction explicitly) (10 min), Check and consolidate: short formative quiz (3 items) including one conceptual question about test vs source charge and one numerical with unit check, then instructor summarises main definitions and units (6 min)
Why this order	This period builds directly on 'Charge Quantisation and Coulomb's Law' by reinterpreting Coulomb force as a field surrounding a source charge; it prepares students to apply the 'Superposition principle for electric field' and to compute flux and dipole behaviour in the next period.

Period 4: Electric Dipole, Dipole Moment and Torque

Concepts	Superposition principle for electric field, Electric flux, Electric dipole, Dipole moment
Objectives	Given a dipole moment p and a uniform electric field E, determine the torque $\tau = p \times E$ and state whether a net force acts on the dipole.
Time allocation	Activate prior knowledge: quick review of electric field vectors and superposition with one worked example using two point charges (show units N/C and m explicitly) (5 min), Introduce new material: define electric dipole and dipole moment $p = q\cdot 2a$ (derive p magnitude and direction from two point charges, carry units C·m), and define electric flux $E\cdot dS$ with

	units $(N/C)\cdot m^2$ (10 min), Demonstration: classroom apparatus or diagram of a dipole in a uniform electric field; show vector forces on charges, measure/compute resulting torque step-by-step to obtain $\tau = p \times E$ with units $N\cdot m$ (8 min), Guided practice: pairs work through two problems — (a) compute p for given q and separation (include units), (b) compute torque magnitude and direction for a dipole at an angle in a uniform E, deriving $\tau = pE \sin\theta$ (10 min), Check for understanding: 3 short conceptual questions (does net force occur? why net charge zero gives no net force in uniform E; compute numerical torque) with quick individual answers and units (4 min), Consolidate and bridge: teacher summarises results, emphasizes derivations and units, and explains how this leads into analysing dipole far-field behaviour and using Gauss's law/Coulomb's law next (3 min)
Why this order	Builds directly on 'Defining and Calculating Electric Field' by using the superposition idea and field vectors to combine point-charge fields into a dipole description; it makes possible calculating the dipole's torque in a uniform field and prepares students for 'Dipole field far-field behaviour, Gauss's law, Coulomb's law'.

Period 5: Dipole Far-Field, Torque and Gauss's Law

Concepts	Dipole field far-field behaviour, Gauss's law, Coulomb's law
Objectives	Given a dipole moment p and a uniform electric field E, determine the torque $\tau = p \times E$ and state whether a net force acts on the dipole.
Time allocation	Activate prior knowledge: brief recall of dipole moment p $(C\cdot m)$, superposition and Coulomb's law with units $(N\cdot m^2/C^2$ or k), and electric flux definition $(V\cdot m$ or $N\cdot m^2/C)$ — quick warm-up example (5 min), Introduce new derivations: step-by-step derivation of dipole far-field E $1/r^3$ using two opposite point charges separated by 2a, carrying units through each step (m, C, N); explicit statement of axis vs equatorial sign (12 min), Demonstration and measurement: teacher demo with two small charged spheres on a jig (charges $\pm q$, separation 2a) showing torque in a uniform external field and qualitative falloff of field with distance; students note measured quantities and units (8 min), Guided problem set: pairs compute torque $\tau = p \times E$ (units $N\cdot m$) for given p and E, and calculate numerical dipole-field values at ra using p $(C\cdot m)$ and ε_0; teacher circulates and derives any algebraic steps requested (10 min), Formative check: short written quiz (2 problems) — (1) state whether net force on ideal dipole in uniform E is zero and explain units, (2) compute ratio of dipole field at two distances; immediate marking and quick feedback (3 min), Consolidation and homework: summarise key derivations (show final formula with units), assign problems linking dipole far-field approximations to upcoming Gauss's-law applications (2 min)
Why this order	This period builds on the prior 'Electric Dipole, Dipole Moment and Torque' ideas of superposition, electric flux and the dipole moment p to derive the dipole's far-field $1/r^3$ behaviour and to compute torque $\tau = p \times E$ with units. It prepares students to apply Gauss's law and far-field approximations when moving to continuous charge densities and the

fields of wires, planes and spherical shells in the next period.

Period 6: Gauss's Law: Fields of Symmetric Charges

Concepts	Continuous charge densities (linear, surface, volume), Superposition principle, Field of an infinitely long uniformly charged wire, Field of an infinite uniformly charged plane, Field of a uniformly charged thin spherical shell
Objectives	Use Gauss's law to derive the electric field of (a) a uniformly charged infinite plane, (b) an infinitely long straight charged wire, and (c) a uniformly charged thin spherical shell for interior and exterior points.
Time allocation	Activate prior knowledge: quick review of Gauss's law statement and units (flux = $Q_{enclosed}/\varepsilon_0$), relate to Coulomb's law with a point charge example (carry units C, m, N/C) (5 min), Derive field of infinite plane: set up Gaussian pillbox, show $dQ = \sigma\, dA$ (σ in C/m^2), compute flux and solve $E = \sigma/(2\varepsilon_0)$, point to a physical example (charged metal plate) and state units N/C (8 min), Derive field of infinite wire: choose cylindrical Gaussian surface, express $dQ = \lambda\, dl$ (λ in C/m), compute flux and solve $E = \lambda/(2\pi \varepsilon_0 r)$, anchor with a charged thin wire example and carry units N/C (8 min), Derive field of thin spherical shell: use spherical Gaussian surface, show $Q = \rho\, dV$ reduces to total q for shell (q in C), derive $E = q/(4\pi \varepsilon_0 r^2)$ outside and E=0 inside, compare to point charge and state units N/C (8 min), Guided practice: teacher-led problem set (pairs) — (a) compute E at 2 cm from λ=5 nC/m wire, (b) E near a σ=10 μC/m^2 plane at 1 mm, (c) E inside/outside shell with q=2 μC at specified r; require showing units at each step (7 min), Check and consolidate: quick formative quiz (3 short calculations and one conceptual question about superposition), discuss common errors and physical interpretation (direction, dependence on r), summarize takeaways (4 min)
Why this order	Builds on Dipole Far-Field, Torque and Gauss's Law by applying Gauss's law and Coulomb's law to continuous charge distributions; completes the module by deriving concrete field formulas for infinite plane, infinite wire, and thin spherical shell which students can use for problem solving and comparisons to point-charge results.

Concepts Taught

Concept	Importance	Summary
Electrostatics	core	Study of forces, fields and potentials arising from static charges.

Concept	Importance	Summary
Everyday examples of static electricity	supporting	Sparks from synthetic clothes, lightning, and shocks from touching charged handles are instances of discharge of static charges.
Two kinds of electric charge and their interaction	core	There are two kinds of electric charge; like charges repel and unlike charges attract each other.
Conductors and insulators	core	Conductors allow electric charge (electrons) to move freely and redistribute charge over their surface; insulators resist passage of charge and retain charge where placed.
Quantisation of electric charge	core	Free charges occur only in integer multiples of a basic unit e (the electron charge).
Coulomb's law	core	The electrostatic force between two point charges varies directly as the product of the charges and inversely as the square of the distance between them; the force acts along the line joining the charges.
Principle of superposition	core	The total electrostatic force on a charge due to multiple charges is the vector sum of the forces due to each charge taken individually; individual pairwise forces are unaffected by the presence of other charges.

Concept	Importance	Summary
Electric field	core	A charge produces an electric field everywhere in the surrounding space; the field at a point gives the force per unit test charge placed there.
Electric field	core	A vector field in space defined by the force per unit test charge; exists at every point and is independent of the test charge used to measure it.
Test charge and source charge	core	A source charge produces the electric field; a test charge is used to probe that field and should be made negligibly small so as not to disturb the source.
Superposition principle for electric field	core	The total electric field at a point due to a system of point charges is the vector sum of fields produced by individual charges.
Electric flux	supporting	Electric flux through an area element is $E \cdot dS = E\, dS \cos\theta$ and represents the (proportional) number of field lines crossing that area element; for closed surfaces dS is taken outward.
Electric dipole	core	An electric dipole is a pair of equal and opposite point charges separated by distance 2a; its total charge is zero but it produces a characteristic field and dipole moment.

Concept	Importance	Summary
Dipole moment	core	The dipole moment vector p has magnitude q times the separation (2a) and direction from the negative to the positive charge: p = q × 2 a.
Dipole field far-field behaviour	core	At distances r >> a, the dipole field falls off as 1/r^3 and can be expressed using the dipole moment p with different signs on axis and equatorial plane.
Gauss's law	core	The total electric flux through any closed surface equals the total charge enclosed divided by epsilon_0.
Coulomb's law	core	The electrostatic force between two point charges is proportional to the product of the charges and inversely proportional to the square of the separation; constant k = 1/(4 pi epsilon_0).
Continuous charge densities (linear, surface, volume)	core	Definitions of macroscopic charge densities: surface density sigma = dQ/dA, linear density lambda = dQ/dl, volume density rho = dQ/dV with SI units C/m^2, C/m, C/m^3 respectively.
Superposition principle	supporting	For an assembly of charges, the net force or field is the vector sum of contributions from individual charges; presence of others does not change pairwise forces.

Concept	Importance	Summary
Field of an infinitely long uniformly charged wire	supporting	For an infinite straight wire with linear charge density lambda, the electric field at radial distance r is E = lambda/(2 pi epsilon_0 r) directed radially.
Field of an infinite uniformly charged plane	supporting	For an infinite plane with surface charge density sigma, the field magnitude is E = sigma/(2 epsilon_0) directed normal to the plane, same on both sides.
Field of a uniformly charged thin spherical shell	supporting	Outside the shell the field equals that of a point charge q at center: E = q/(4 pi epsilon_0 r^2); inside the shell E = 0.

Formulae

Quantisation of charge

q = n e

where q = total charge on a body (C); n = integer (positive or negative); e = basic unit of charge (magnitude of electron charge) (C)

Coulomb's law (scalar)

F = (1 / (4 pi epsilon_0)) * (q1 q2 / r^2)

where F = magnitude of the electrostatic force between two point charges (N); q_1, q_2 = the two point charges (C); r = distance between the two charges (m); \epsilon_0 = permittivity of free space (C^2 N^-1 m^-2)

Coulomb's law (vector)

F_12 = (1 / (4 pi epsilon_0)) * (q1 q2 / r_21^2) * rhat_21

where F_12 = force on charge 1 due to charge 2 (vector) (N); q_1, q_2 = point charges 1 and 2 (C); r_{21} = distance from charge 2 to charge 1 (m); \hat{r}_{21} = unit vector from charge 2 to charge 1; \epsilon_0 = permittivity of free space (C^2 N^-1 m^-2)

Force on a charge in an electric field

F(r) = q E(r)

where F = Force on the test charge (N); q = Test charge (C); E = Electric field at r (N C^{-1})

Electric field of a point charge (element)

E = (1/(4 pi epsilon_0)) (q / r^2) r̂

where q = Source point charge (C); r = Distance from source to field point (m); r̂ = Unit vector from source to field point; epsilon_0 = Permittivity of free space (C^2 N^{-1} m^{-2})

Superposition sum for fields

E(r) = sum_i [(1/(4 pi epsilon_0)) (q_i / r_{iP}^2) r̂_{iP}]

where q_i = i-th source charge (C); r_{iP} = Distance from i-th charge to point P (m); r̂_{iP} = Unit vector from i-th charge to P

Electric flux through area element

dPhi = E · dS = E dS cos theta

where E = Electric field (N C^{-1}); dS = Area element vector (magnitude dS and direction normal) (m^2); \theta = Angle between E and dS (rad)

Dipole moment definition

p = q × 2 a

where p = Dipole moment vector (C m); q = Magnitude of each charge (C); 2a = Separation between charges (m)

Far-field dipole axis formula

E (on axis, r>>a) = (1/(4 pi epsilon_0)) (2 p / r^3)

where p = Dipole moment (C m); r = Distance from dipole centre (m)

Far-field dipole equatorial formula

E (on equatorial plane, r>>a) = - (1/(4 pi epsilon_0)) (p / r^3)

where p = Dipole moment (C m); r = Distance from dipole centre (m)

Gauss's law (integral)

Flux through closed surface = enclosed charge / epsilon_0

where Phi = electric flux through closed surface (N m^2 C^-1); E = electric field vector (N C^-1); dS = vector area element (outward normal) (m^2); q_enc = total charge enclosed by surface S (C); epsilon_0 = permittivity of free space (N^-1 m^-2 C^2)

Coulomb's law (magnitude/vector)

Force between two point charges = (1/4pi epsilon_0) * q1 q2 / r^2 in the radial direction

where F = electrostatic force on one charge due to the other (N); q1,q2 = point charges (C); r = separation between charges (m); hat{r} = unit vector from one charge to the other; epsilon_0 = permittivity of free space (N^-1 m^-2 C^2)

Field of infinite line charge

Electric field at distance r from infinite line charge = lambda/(2 pi epsilon_0 r) radially

where E = electric field magnitude (N C^-1); lambda = linear charge density (C m^-1); r = radial distance from wire (m)

Field of infinite plane sheet

Electric field due to infinite plane sheet = sigma/(2 epsilon_0) normal to sheet

where E = electric field magnitude (N C^-1); sigma = surface charge density (C m^-2)

Field of uniformly charged spherical shell (outside/inside)

Outside: E = q/(4 pi epsilon_0 r^2); Inside: E = 0 (for r<R)

where q = total charge on shell (C); r = distance from centre (m); epsilon_0 = permittivity of free space (N^-1 m^-2 C^2)